HelSpin — User Manual

Version 0.5.17

Compare Bruker NMR spectra and build publication figures.

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1. What HelSpin is for

You have a folder of Bruker datasets and a question that needs several of them side by side: did the reaction go, does this batch match the reference, which of six conditions gave the cleanest product. HelSpin opens them together, lets you line them up, and exports the result at publication quality.

It reads Bruker data directly from wherever it already lives, including a mounted spectrometer share. **It never writes to your data.** Everything you adjust — scale, position, colour, zoom — is display state, held in HelSpin and saved in its own session files.

2. Installing

On macOS, if you have `HelSpin-<version>-<arch>.dmg`: open it, drag HelSpin to Applications, and on first launch **right-click the app and choose Open** to get past Gatekeeper (once only — the app is not signed with an Apple Developer ID). No Python or conda is needed. Check the architecture in the filename: `arm64` for Apple Silicon, `x86_64` for Intel; they are not interchangeable.

Otherwise, and on Windows and Linux, see `INSTALL.md`. Linux needs one system library (`libxcb-cursor0`) before the first run.

3. Getting started

Adding a data root

File → **Add Data Root…**, then pick the folder your datasets sit under. Point it at a directory that *contains* sample folders, not at a single dataset. HelSpin walks down looking for the Bruker structure (a sample folder holding numbered experiment folders) and shows what it finds.

You can add more than one root — a network share and a local archive, say — and they persist between sessions.

Symlinks are followed, so a root made of links to the real instrument mounts works normally. Links pointing back at a parent are detected and not followed twice, so a loop cannot hang the scan.

The browser

The left panel is the tree: root → sample → experiment. Type in the filter box to narrow by sample name or pulse programme; the filter reaches samples that have not been expanded yet.

The tree does **not** watch the filesystem. Network shares make that unreliable, so refreshing is explicit: right-click a node to refresh it, or **Refresh All** / **F5**.

Opening spectra

Drag a dataset onto the canvas, or double-click it. Repeat to add more. Each one becomes a *trace* listed in the right-hand panel.

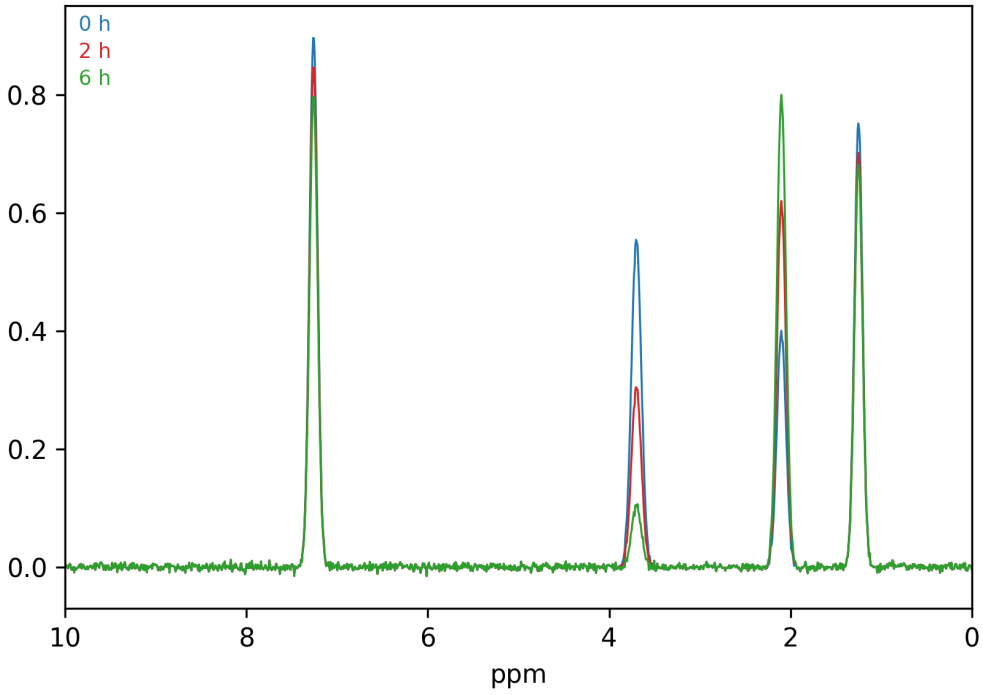
If a dataset does not appear, ask HelSpin what it sees on disk:

```
helspin --check /path/to/SAMPLE
```

That reports, experiment by experiment, whether a plottable spectrum is actually there.

4. Arrangements: overlay and stacked

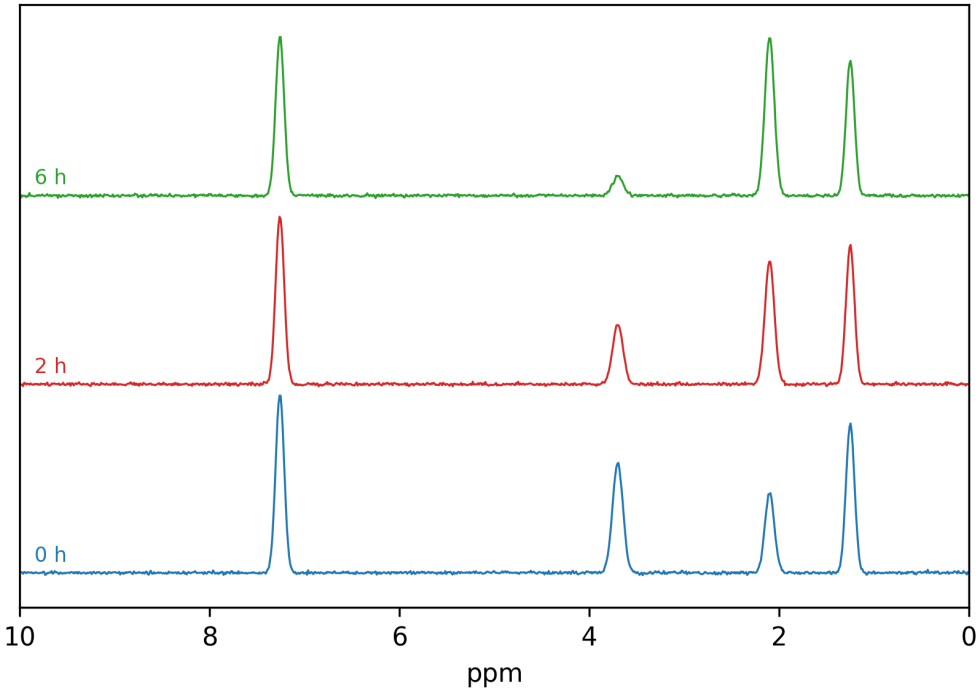
Overlay draws every spectrum on one set of axes, sharing a baseline. Use it to compare peak positions and relative intensities directly.



Overlay

Stacked gives each spectrum its own lane, evenly spaced. Use it when the traces would otherwise obscure each other, and for showing a progression.

In a stack each spectrum's name sits **beside its own trace**, so a reader does not have to match colours to work out which is which. The name is anchored to the lane's baseline, so it stays put when you scale or magnify the spectrum, and follows the lane when you zoom. Drag a name to reposition it; **Reset label positions** puts them back.



Stacked

Switch with the arrangement control. The frame is re-fitted on switching, because offsets that suit one layout rarely suit the other.

5. Adjusting a spectrum

Select a trace in the right-hand panel first — adjustments apply to the selection.

Y scale — the wheel

With both zoom toggles off, the **mouse wheel over the plot scales the selected spectrum** vertically. This changes that spectrum's height relative to the others, which is what you want when one sample is far more concentrated than another.

Nothing happens if no trace is selected. That is deliberate: guessing which spectrum you meant would be worse than asking you to click one.

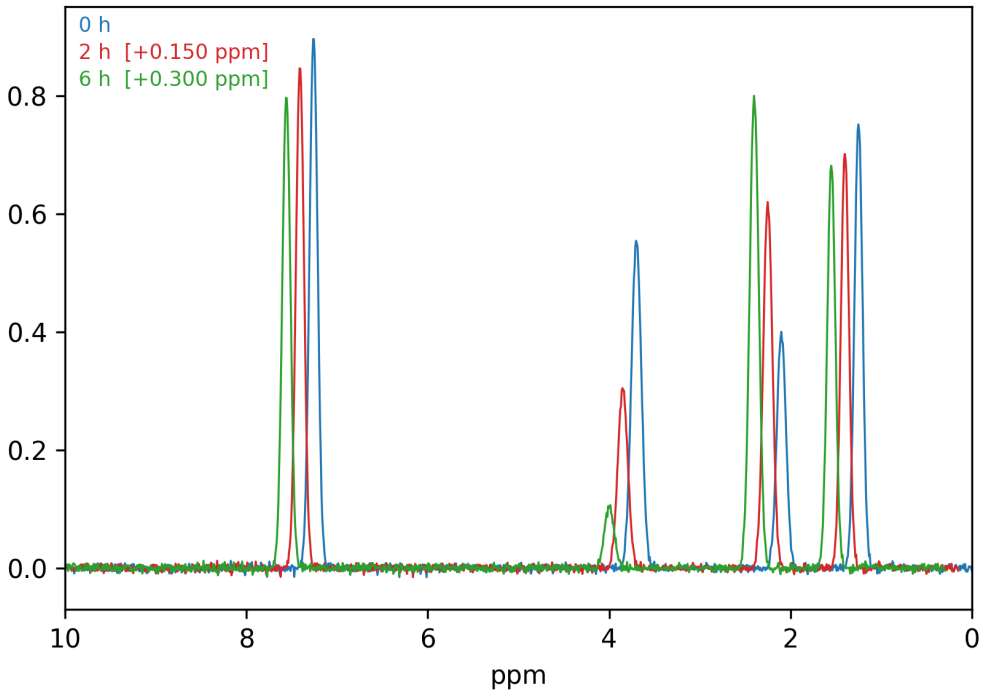
Y offset

Moves the selected spectrum up or down. Pure translation — it does not rescale anything and does not re-fit the view.

To bottom drops the selected spectrum to the baseline; **Bottom all** does every spectrum at once.

X offset — shifting along ppm

Spectrum list → **X offset**, in ppm. Shifts the selected spectrum horizontally. Two uses: correcting spectra that were referenced slightly differently, and skewing a stack into a cascade.



X offset

The step size and range come from the spectrum's own width, so the control behaves sensibly for a ^{1}H spectrum spanning 12 ppm and a ^{13}C one spanning 200. You can shift up to half a spectrum width in either direction.

> **A shifted spectrum says so.** An X offset moves a trace along the chemical shift axis, so its peaks no longer read at their true values. HelSpin marks any shifted trace in the spectrum list *and on the plot label* — `2 h [+0.150 ppm]` — so a figure never quietly claims a peak sits somewhere it does not. The marker disappears when the shift returns to zero. The underlying data is never modified. Peak readout always reports true, unshifted ppm, and **Clear X offsets** returns everything to its real chemical shift in one action.

Appearance

Colour, line width and line style are per-spectrum, in the spectrum list. Defaults come from the slot palette in Preferences, so the first spectrum you open always looks the same way.

6. Zooming

Two independent toggles in the bottom bar: **X zoom** and **Y zoom**. They are not exclusive — with both on, the wheel zooms both axes at once.

Toggle	What the wheel does
Both off	Scales the **selected spectrum** vertically
X zoom	Zooms the ppm axis about the cursor
Y zoom (overlay)	Zooms the intensity axis about the cursor
Y zoom (stacked)	**Magnifies all spectra**, leaving the lanes where they are
Both on	Zooms both axes together

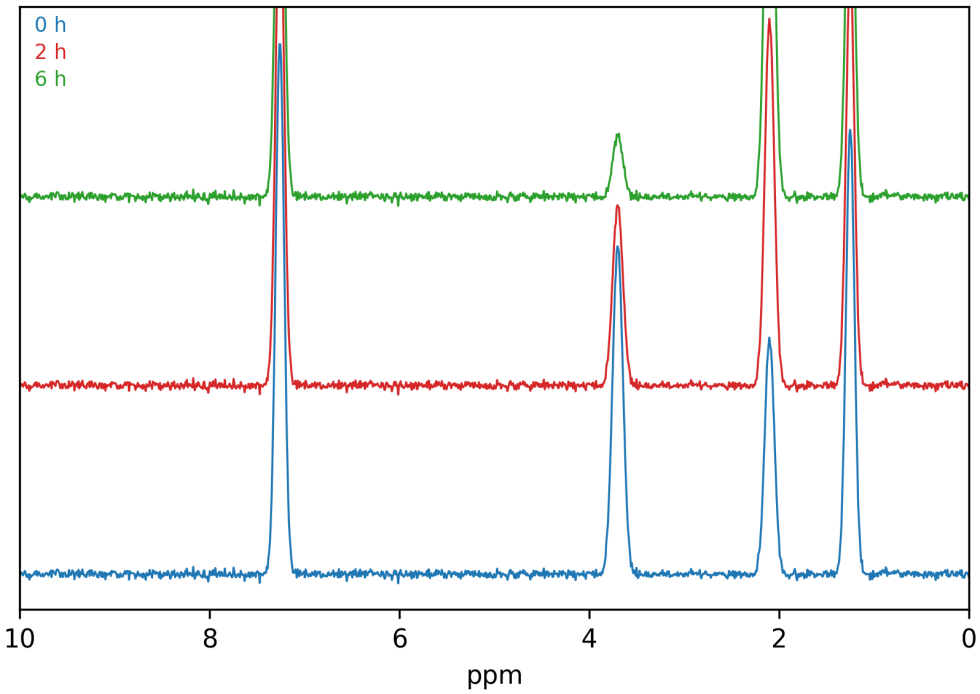
Zooming keeps the point under the cursor fixed, so a peak of interest does not walk off the edge.

With **X zoom** on, you can also drag a box on the plot to zoom into it.

Y zoom in stacked mode is different, on purpose

In a stack, the layout *is* a set of lanes at fixed baselines. Narrowing the vertical window there would not magnify the stack — it would crop it, pushing whole spectra off the canvas.

So in stacked mode the wheel **magnifies the traces** and leaves the baselines exactly where they are. Peaks grow in place, overflow their lane, and can run off the top of the canvas.



Stacked and magnified

That overflow is intended. Bringing up a weak signal beside a strong one can need magnification of many orders of magnitude, so **there is no limit** in either direction.

Getting back

- **Fit Y** — re-frames the vertical axis and clears any Y zoom or magnification.
- **Full** — shows everything, in every axis.
- **Ctrl+Z** — undoes it, including Fit Y.

A zoom is sticky: it survives loading another spectrum, removing one, changing the ppm window and redrawing. Turning the toggle off does not undo it. Only Fit Y, Full, reset or undo clear it.

7. Setting the ppm window

The bottom bar takes a typed range. ppm axes descend, so the **left** box is the higher value. **Apply** sets it; **Full** clears it.

Recently used ranges are remembered **between runs**, so returning to a window you use often is one click. The list starts seeded with three common windows rather than empty:

Range	Typical use
0 to 12 ppm	the standard ^{1}H sweep
-1 to 13 ppm	the same, with a margin at each end
5 to 13 ppm	downfield only: aromatics, amides

Ranges you apply are added to the front, so the list becomes yours with use.

Narrowing the window re-fits the vertical frame to the data actually in view — otherwise zooming into a quiet region would keep the scale set by peaks outside it and show you a flat line. An explicit Y zoom outranks that and is kept.

8. Undo

Ctrl+Z / **Ctrl+Y** (or **Ctrl+Shift+Z**). Forty steps.

Undo covers display state — scales, offsets, zooms, colours, adding and removing spectra. It never re-reads a file, so undoing a change costs nothing even with data on a slow share, and undoing a removal brings the spectrum back instantly with its data intact.

Typing into a spin box is one undo step, not one per keystroke.

9. Sessions

File → **Save Session…** writes everything about the current view: which datasets are open, and every scale, offset, colour, zoom and window. **File** → **Open Session…** restores it.

Sessions store *paths*, not spectra. If the underlying data has moved, the session cannot find it — keep the data root stable, or re-add the root.

Restoring clears the undo history, so an opened session cannot be undone into the state that preceded it.

10. Exporting a figure

Save Image…. Choose PNG, PDF, or SVG by the extension you type; PDF and SVG stay vector and are what you want for a manuscript.

- **300 dpi** by default.
- **Transparent background** is available for slides.
- The exported figure matches what is on screen, including any X offset markers.

11. Where HelSpin keeps its files

What	macOS / Linux	Windows
Settings	`~/.config/HelSpin/`	registry
Index cache, licence	`~/.cache/helspin/`	`%LOCALAPPDATA%\HelSpin\cache`

On Linux, `XDG_CACHE_HOME` is honoured — worth setting if `/home` is a slow network mount, since the cache belongs on local disk. `HELSPIN_CACHE_DIR` overrides it explicitly on any platform.

Deleting the cache is safe; it is rebuilt on the next scan.

12. Troubleshooting

A sample is missing from the tree. Run `helspin --check /path/to/SAMPLE`. Note that Bruker writes `acqus`, `pdata` and `1r` in lower case and Linux filesystems are case-sensitive, so data copied from Windows by a tool that changed case will not be recognised.

Permission denied on a share. Confirm you can read it outside HelSpin first (`ls /mnt/nmr`). An unreadable subtree is stepped over silently by design, so one locked folder does not cost you the rest of the share.

Linux: the application will not start, with a Qt platform plugin error. Install `libxcb-cursor0` (Debian/Ubuntu), `xcb-util-cursor` (Fedora/Arch). See INSTALL.md.

Linux: the window never appears, no error. Check for a leftover `QT_QPA_PLATFORM=offscreen` in your shell.

A spectrum vanished after zooming. Press **Fit Y**.

The plot looks wrong and you cannot say why. Full, then **Fit Y**, returns to a neutral view without closing anything.

13. Known limitations

Stated plainly, because finding these out mid-figure is worse.

- **Differences and sums ignore an X offset.** If you align two spectra horizontally and then subtract, the subtraction uses their true ppm axes, not the aligned ones.
- **No automatic file-watching.** Refresh is explicit, by design — see §2.
- **The TopSpin bridge is not wired up.** TopSpin identifiers parse, but there is no live connection to a running TopSpin.
- **Windows and macOS are not yet verified end to end.** Development and the test suite currently run on Linux. The Windows installer path in particular has not been exercised from start to finish.

14. Licence

Free for academic research, teaching and personal use. Commercial use requires a licence — enquiries to **iwai@ligsciss.com**. Full terms in `LICENSE`, and readable in the application at **Help** → **Licence**.

Qt (via PySide6) is LGPLv3 and is packaged one-directory so its shared libraries stay replaceable. nmrglue, NumPy and SciPy are BSD; matplotlib is BSD-style.

Markers

Right-click the spectrum canvas to pin a vertical or horizontal marker where you clicked, labelled with its value. **Remove nearest marker** drops the closest one; **Clear all markers** empties them.

Markers are held in data coordinates, so they stay on the feature when you pan, zoom or rescale. Their colour, width, dash and opacity are in Preferences, and default to matching the cursor crosshair.

CEST mode

Toolbar **CEST…**, or File → CEST…. Opens its own window. If exactly one experiment is selected in the browser it is loaded straight away.

The panel handles both Waudby-lab pseudo-2D experiments and picks the tab from PULPROG: `19f_calib_nut` opens the calibration tab, `19f_cest` the Z-spectrum tab.

Show fitted curves (in the header) hides the fitted lines on both tabs and leaves the measured points, residuals and reported numbers. Useful for judging a marginal dip against the data rather than against the model drawn through it.

Which data it reads

Raw `ser` by default. `xf2` stores SI(F1) rows, and TopSpin defaults SI to a power of two — so a 42-offset series processed at SI(F1) = 32 silently loses ten offsets. The resulting `2rr` is a valid file with no record of the loss. The panel treats the frequency list as the authority on how many rows should exist, and says so in the notes box when pdata disagrees.

Set **Source** to *Processed 2rr* if you want the phasing you set in TopSpin; the panel will still warn if rows are missing. SI(F1) larger than TD(F1) is fine — the surplus rows are blank padding and are trimmed.

To keep TopSpin's own view complete, set `1 SI` to the next power of two above your offset count (64 for 42 offsets) before `xf2`, and accept the blank trailing rows.

Calibration tab

Choose a peak centre and half-width, then **Fit nutation**. The window is symmetric — centre ± half-width, so 0.03 means a 0.06 ppm span — and the **Window** readout shows the resulting range and its point count. It is applied identically to every row and the result stays signed, because the sign inversion across rows is the measurement.

- **Measurement** — integration is the default and the right choice here. A narrow window works best: on 19F test data, residuals fell from 7.0% at ±0.20 ppm to 3.9% at ±0.03 ppm.
- **Model** — *Automatic* compares a plain sine against a damped one on AICc and reports which won and by how much.
- **Target field** — the field you want. It need not match this calibration's CNST8.

Results end with a **TO SET ON THE SPECTROMETER** table giving the `CNST` value to type for each true field, with the pulse length and power the sequence will derive from it. Lowering `CNST` is the whole adjustment, and in `19f_cest` it is the only one available — that sequence recomputes `plw25` unconditionally, so anything typed into `PLW25` is overwritten when the pulse program compiles. Record the true field in the title, since acqus will show the lowered `CNST` rather than the field you applied.

Results also give the fitted B1 with an error bar, T2 effective, residuals, and two power figures kept deliberately apart: the **correction factor** `(nominal/fitted)`^{2}, which multiplies any nominal power on the same probe and tuning and so transfers to a CEST experiment at a different field; and the absolute power for your target field, which also changes the field and therefore does not transfer. **Export record…** writes a record sheet matching Part 8 of the 19F CEST manual.

Z-spectrum tab

Saturation offsets are read from the experiment's own `lists/f1/` copy of the FQ1LIST. Bare Hz, ppm (`P`), absolute MHz (`SFO`) and `BF` lists are all understood, as are comments and Windows line endings. Without a list the panel says so rather than plotting against row number. A series with fewer rows than offsets is plotted against the offsets it has, with the missing range named; more rows than offsets is refused, since the extras cannot be matched to anything.

- **Measurement** — peak height is the default and usually correct. On 19F test data, repeated reference offsets scattered by 0.007 using height against 0.090 using the integral: a saturated peak leaves a drifting baseline that an integral accumulates without bound, far enough to drive I/I■ negative.
- **Reference from** — offsets at least this far out are averaged to give I■. *Automatic* picks the remote ones. Two or more give a scatter figure, which is a direct measure of the noise on every other point — include a couple of remote offsets in your list for this reason.
- **Fit deepest dip** — a Lorentzian on the points near the dip, reporting its centre in Hz and ppm from the carrier. A dip sitting on the carrier is flagged as likely direct saturation rather than exchange. Every other minimum clearing the sigma threshold is listed with its significance.
- **Error bars from spectrum noise** — one sigma per point, anchored to the spread of the repeated I0 references. A shaded band marks ±1 sigma.
- **Subtract fitted dip** — removes the fitted profile so direct-saturation wings stop hiding a smaller dip beside them. For locating a feature, not for quantifying it.
- **Fit two dips together** — fits the deepest dip and the best other candidate simultaneously, so the major dip's wings do not bias the minor depth. Reports both positions, their separation, and the minor dip's share of total depth. That share is a **depth ratio, not a population**: dip depth also depends on exchange rate, saturation field, `D18` and relaxation. Comparable between experiments run under identical conditions; not a substitute for Bloch-McConnell.
- **Zooming** — scroll to zoom X about the cursor, shift-scroll for Y, double-click to reset. The range boxes are a readout and follow the wheel; **Full range** also resets. The direct-saturation dip runs to near zero, so the baseline where a bound-state dip would sit is compressed — zooming Y to roughly 0.95–1.02 is usually what makes one visible.
- **Keep on plot / Remove / Clear all** — overlay several Z-spectra for comparison. Each keeps its own I■ normalisation, so experiments with different gain, scans or concentration remain comparable. Entries are labelled with the experiment, saturation field and D18. Use this for the power series (30/60/100 Hz) and the partner-free control from Part 7 of the 19F CEST manual — a real exchange feature moves predictably with field and is absent from the control, and both are far easier to judge on one axis.
- **Add experiment…** — overlay another CEST expno without replacing the loaded one. It is measured with the current half-width, measurement, reference and error-bar settings, on the peak nearest the current main peak. Selecting several expnos in the browser before opening CEST loads the first and overlays the rest in one step.

- **Save image…** (Ctrl+S / Cmd+S, or right-click the plot) — saves the plot exactly as zoomed, in the same formats as the main canvas: PNG, JPEG, TIFF, SVG, PDF, EPS, PS or PowerPoint (300 dpi for raster).
- **Right-click menu** — as on the main canvas: Save image, Full range, and vertical/horizontal markers (add at the click, remove nearest, clear all), styled from Preferences. The Z-spectrum menu also adds or keeps overlays.
- **Drag and drop** — drag expnos from the browser onto the CEST window to overlay their Z-spectra.

Scroll the wheel over the plot to zoom x about the cursor; hold Shift to zoom y. The range boxes follow.

Working with several resonances

Find peak picks the resonance nearest the carrier — the one the saturation offsets are measured from, and normally the one you want. The **peak list** below shows everything found, with each peak's offset from the carrier. The main peak is marked and always measured; tick another to add its Z-spectrum to the plot as a dashed curve, or double-click to make it the main peak instead. **Peak centre** stays freely editable if the detector misses one.

Note that the carrier and the observed peak are different things. The peak is where you measure intensity; the carrier is the zero of the saturation axis. If they differ, the direct-saturation dip appears off-centre — a 0.137 ppm difference put it at +78 Hz on the test data. Setting O1P exactly on the resonance avoids this.

detect above N x noise controls how many peaks are listed. There is no universal setting: on one sample the main peak was 55x noise and a plausible second resonance only 6x. Lower it to look for weak resonances, raise it when the list fills with baseline ripple.

Hunting a bound-state dip

1. Shift-scroll to about 0.95–1.02 in y so the baseline fills the plot.
2. Tick **Subtract fitted dip** to flatten the direct-saturation wings.
3. Read the candidate list. Three sigma is worth following up, five is convincing.
4. Confirm it: a real exchange feature persists in a repeat, moves predictably with saturation field, and is absent from a partner-free control. Direct saturation sits on the carrier; a bound state does not.

If nothing clears the threshold, the limit is noise rather than the tool. More scans, longer `D18`, or higher concentration lower the sigma figure, and the candidate list will follow.

This locates dips; it does not measure exchange. Turning a centre and depth into kex and a populated fraction needs Bloch-McConnell fitting against several saturation fields, which is not attempted.